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Grapevine double cropping: Theoretical and applied aspects for year-round production of grapes

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ABSTRACT

Dormancy is a critical stage for the development of floral organs in grapevine buds. In regions with insufficient low temperatures, the dormancy period may be prolonged, leading to delayed germination, irregular budbreak, and reduced yields. This indicates the imperative to formulate scientific and sustainable strategies to effectively address grape dormancy and enhance early budbreak quality, while also undertaking comprehensive studies on the critical factors affecting secondary inflorescence formation, thus ensuring consistent yield and quality in grape cultivation systems. This review summarizes recent advances in understanding stress-induced secondary flowering in grapevine, with a focus on postharvest practices like leaf-degreening, pruning, and hydrogen cyanamide application, as well as key mechanisms involving gene regulation, ROS signaling, hormonal crosstalk, and carbohydrate metabolism. Among these practices, leaf degreening promotes nutrient backflow and the differentiation of secondary flower buds; pruning stimulates budburst and increases the proportion of floral buds; and hydrogen cyanamide (HC) application improves the uniformity of budbreak while enhancing both the budbreak and flowering rates in grapevines. Incorporating a variety of grape varieties further facilitates year-round cultivation, thereby presenting a potential pathway for sustainable production. In conclusion, the review draws attention to current challenges and provides insights into future applications, to optimize viticultural practices in response to evolving market and environmental demands.

1. Introduction

Grapevine, one of the earliest domesticated fruit crops, holds significant economic importance with the global grape cultivation area of 6.7 million hectares in 2022, and global grape production of 75 million tons (Topcu and Başer, 2025). Today, grape farming spans six continents, thriving mostly in temperate areas defined by warm, dry summers and mild winters (Dong et al., 2023; Tonietto and Carbonneau, 2004). Environmental conditions have a substantial effect on flowering and fruiting (Dinkara et al., 2024; Mitra et al., 2018).

In grapevines, practices such as pruning, the application of degreening chemicals, or the exogenous application of hydrogen cyanamide (HC) effectively regulating the physiological state of the plant, thereby promoting growth, inducing secondary flowering, and maximizing yield (Ionescu et al., 2017; Zhang et al., 2023). Specifically, such practices act as forms of abiotic stress, triggering specific signaling pathways and transcription factors that enhance ROS signaling, modulate hormone levels (e.g., decreasing ABA and increasing GA and ethylene), and activate key floral integrator genes, thus facilitating the plant's transition from dormancy to flowering and ensuring a complete life cycle (Balazadeh et al., 2012; Han et al., 2024; Li et al., 2024).

This physiological flexibility has important implications for double cropping systems, where grapevines are induced to flower and fruit twice per year. Traditional grape cultivation typically involves a single annual harvest. However, recent advancements have enabled the implementation of a double-cropping system for some cultivars by exploiting the plant's responsiveness to environmental cues (Qiu et al., 2019). The implementation of a two-crop cultivation system annually can regulate the harvest period of grapes, enhance the quality of fruits, and enhance economic benefits. Successful examples of two-crop

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agriculture have been documented in locations including Taiwan, Fujian, Yunnan, Guangxi, and Hunan in China, as well as in other nations like as Brazil (Ahmed et al., 2019; Favero et al., 2011; Koyama et al., 2020; Li et al., 2023; Poni et al., 2021; Tian et al., 2019).

This review focuses on the regulation of dormancy and flowering induction in double-cropping systems, aiming to bridge the gap between environmental influences, physiological regulation, and the optimization of double-cropping practices. It also outlines innovative strategies, including the development of non-overlapping double-cropping systems and the careful selection of cultivars, to enhance the feasibility and sustainability of year-round grape production.

2. Physiological basis of secondary flowering of the grapevine induced by stress

The most important step of grape flowering is floral induction, which is a complex developmental process that involves floral initiation, floral differentiation, and the final blooming stage. The precise timing of blossoming is guaranteed by a complex network of endogenous hormones and environmental factors that regulate floral induction (Noyce et al., 2015). Grapevines have physiological plasticity, which allows them to adapt to abiotic challenges such as changing climate conditions and water management strategies. This adaptation permits grapevines to blossom and yield fruit twice a year under specific conditions, demonstrating substantial environmental resilience (Monteiro et al., 2021). Stress-induced physiological reprogramming enables grapevines to carefully regulate flowering time by balancing internal hormones and external environmental signals, thereby achieving maximum reproductive success across diverse climates.

2.1. Signaling role of reactive oxygen species (ROS) in floral development

Hydrogen peroxide (H_2O_2), as a type of reactive oxygen species (ROS), influences flowering time and floral organ development (Balazadeh et al., 2012). Buds subjected to pruning and HC treatment produced significantly higher levels of reactive oxygen species (ROS) and nitric oxide (NO) (Sudawan et al. 2016). Following HC application, mitochondrial respiratory disruption causes transitory oxidative stress, which is characterised by increased ROS generation, decreased tricarboxylic acid (TCA) cycle activity, reduced ATP synthesis, and increased ethylene biosynthesis (Sudawan et al., 2016). Hydrogen cyanamide treatment disrupts mitochondrial function, thereby inducing the accumulation of reactive oxygen species (ROS) and sustaining the activation of growth hormone signaling pathways (Fan et al., 2023). HC treatment activates RBOH proteins, a class of multimeric flavocytochrome complexes, which catalyze the reduction of molecular oxygen (O_2) to superoxide radicals (O_2^-) through electron transport chain activity, ultimately ensures initiating ROS signaling (Tognetti et al., 2012). The resulting accumulation of ROS not only upregulates the expression of scavenging-related genes such as *VvPOD3, VvPOD29*, and *VvGSH-Px* (Liang et al., 2019), but also activates ROS signaling pathways in *Camellia oleifera* floral buds, ultimately regulating cell wall expansion and controlling flowering time (Lin et al., 2022).

2.2. Hormonal crosstalk in secondary inflorescence regulation

Stress-induced physiological remodeling underlies the regulation of secondary flower production in grapevines, with plant growth regulators modulating this process through complex signaling networks. In grapevine, gibberellic acid (GA) serves as a physiological link between flowering-related genes and other phytohormone signals (Han et al., 2024). Recent research reveals the involvement of HC in GA metabolism to promote bud break in *Vitis vinifera*. In early floral buds of 'Red Globe' grapevines treated with HC, the expression of GA biosynthesis-related genes (*VvGA3ox1, VvGA3ox2, VvGA20ox3*, and *VvGA20ox6*) was suppressed, whereas the expression of the GA catabolic gene (*VvGA2ox3*) was upregulated (Li et al., 2024). The regulation of *VvGID1B, VvDELLA* and *VvPIF3* expression by HC treatment resulted in the induction of GA anabolic-related genes and the stimulation of bud breakage (Liang et al., 2019). Besides, HC induced an up-regulation of *VvGA2ox* expression and a down-regulation of *VvGA3ox* and *VvGA20ox* expression, resulting in the decrease of GA_1 expression level (Zheng et al., 2018b).

In the Indole-3-acetic acid (IAA) signal transduction pathway, HC treatment resulted in the downregulation of *VvAUX3* and *VvAUX5* between 12 and 16 days, while simultaneously upregulating the expression of response factors directly associated with plant growth within the IAA pathway, such as *VvARF5, VvIAA6, VvIAA29, VvGH3*, and *VvARG7* (Liang et al., 2019). HC has been found to influence abscisic acid (ABA) homeostasis by downregulating the transcription of important ABA biosynthesis genes, such as *VvNCED* and *VvERICO*, therefore lowering endogenous ABA levels (Zheng et al., 2018a).

Jasmonic acid (JA) affects floral organ development and advances flower opening (Browse and Wallis, 2019). In blueberry (*Vaccinium corymbosum* L. interspecific hybrids) flower buds subjected to HC, the production of JA persisted at a comparatively elevated level, even when distinct individual flowers within the inflorescence were identifiable. The temporary reduction in ABA levels in the flower buds may be influenced by the accumulation of JA (Lin and Agehara, 2021). The similar findings was found in sweet cherry (*Prunus avium* L.) flower buds with either 2 % HC or water (control) that HC affects floral development in sweet cherries via the jasmonate signalling pathway, which is regulated by the MYB21/MYB108 transcription factors (Ionescu et al., 2017).

HC promotes bud break in grapevines through coordinated activation of Cytokinin (CTK) and ethylene signaling pathways. In the CTK signal transduction pathway, the expression of *VvHPK4, VvHPT4* and *VvARR6* was found to be upregulated during HC-induced germination (Liang et al., 2019). In blueberries, the initial development of flower buds following hydrogen cyanamide treatment is most likely driven by CTK rather than IAA (Lin and Agehara, 2021). In perennial species like grapevine (*Vitis vinifera* L.), ethylene plays a crucial role in releasing bud dormancy and facilitating flowering (Liang et al., 2019). Recent evidence points to molecular connections between HC and ethylene signalling pathways, with ethylene response factors (ERFs) identified as critical regulatory hubs that regulate flowering timing in perennial plants (Ophir et al., 2009). Ethylene influences ABA breakdown and grape bud germination by altering ABA signaling factors, diminishing the expression of *VvXERICO* and *VvNCED* genes, and enhancing the expression of *VvABA8'OH* homologs (Zheng et al., 2015). Concurrently, HC stimulates ethylene biosynthesis by inducing the expression of ethylene synthesis genes *VvACS* (1-aminocyclopropane-1-carboxylate synthase) and *VvACO* (ACC oxidase), which collectively drive dormancy release through synergistic hormonal crosstalk (Shi et al., 2018).

2.3. Regulation of carbohydrate metabolism during floral transition

Beyond ROS signaling and hormonal regulation, stress-induced physiological adjustments—particularly the mobilization and allocation of carbohydrate reserves—play a pivotal role in promoting flowering and secondary inflorescence development in grapevines (Lebon et al., 2005). Previous research has found that carbohydrate reserves in grapevines are primarily stored in the form of starch and soluble sugars (Vaillant-Gaveau et al., 2014). Developing inflorescences primarily acquire sugars, including sucrose, glucose, and fructose, from xylem and phloem sap (Campbell and Strother, 1996; Lebon, et al., 2008). Genes encoding hexose or sucrose transporters are expressed in grapevine floral organs, among which the *VvSuc27* gene, involved in the sucrose/H^+ co-transport process, exhibits predominant expression in flowers (Davies et al., 1999). Grape inflorescences may adjust to varying degrees of low-temperature stress via distinct carbon metabolism mechanisms: mild chilling (4 °C) stimulates the soluble sugar synthesis route, whereas freezing (≤0 °C) concurrently activates the starch metabolism pathway

(Sawicki et al., 2012). In grapevine, upon the termination of the dormancy period, this carbohydrate conversion process is reversed, with starch levels increasing and soluble sugar levels decreasing (Călugăr et al., 2019). These carbohydrate accumulation and metabolism processes are crucial for ensuring normal grape flowering. Moreover, Carbohydrates act as signaling molecules that regulate numerous flowering-associated genes (Gibson, 2005; Ohto, et al., 2001; Wu et al., 2025). In grapevine (*Vitis vinifera*), the *VFL* gene is a key regulator of reproductive development, promoting early inflorescence and floral organ formation (Carmona et al., 2002). Notably, sucrose signaling specifically upregulates *VFL* expression at inflorescence branching points, linking carbohydrate status to floral organogenesis (Yang et al., 2011). During flower bud differentiation of winter buds in 'Sunshine Muscat' following hydrogen cyanamide (HC) treatment, correlation analysis revealed positive associations of *VvAP1* and *VvLFY* with soluble sugars and starch, whereas *VvSOC1* was negatively correlated with soluble sugar changes (Zhu, 2020). HC treatment participates in bud dormancy release by inducing α-amylase activity and promoting sugar accumulation, thereby effectively accelerating flower bud break in single - node cuttings and significantly enhancing the metabolic turnover rate of soluble sugars within the buds (Mohamed et al., 2012). This conversion of starch to soluble sugars, propelled by elevated amylase activity, is paramount for fulfilling the metabolic demands associated with sprouting and efficaciously facilitating dormancy release. Following HC treatment, the starch content in grape buds exhibited a continuous decrease from 0 to 16 days, whereas the soluble sugar content showed a rapid increase from 0 to 8 days, followed by a significant decline. The modification in carbohydrate metabolism coincided with the up-regulation of *VvAM* and *VvBM* genes linked to starch degradation, which collectively enhanced the soluble sugar content (Liang et al., 2019).

Based on relevant literature, we propose a working model in which HC treatment, as a stress-induced stimulus, triggers physiological reprogramming that underlies dormancy release and secondary flowering in grapevines (Fig. 1). HC treatment decreased ABA levels by upregulating ABA degrading enzymes, induced GA anabolic gene expression, and increased ethylene and CTK levels. This hormone balance triggers changes in gene expression that activate genes associated with growth and development, promoting germination and growth. HC also induced rapid degradation of starch in grape bud, increased soluble sugar content, and provided energy and carbon source for bud germination. These hormonal and carbohydrate dynamics are tightly integrated with redox signaling: reactive oxygen species (ROS) function as secondary messengers that modulate hormone biosynthesis and signal transduction. Collectively, the crosstalk among ROS, hormonal, and carbohydrate pathways establishes a dynamic regulatory network that maintains redox and metabolic homeostasis, facilitating dormancy release, growth resumption, and timely flowering in grapevine.

2.4. Transcriptional control of secondary inflorescence initiation

Floral induction in grapevine is a stress-triggered physiological process that elicits a comprehensive reprogramming of physiological states accompanied by widespread changes in gene expression. This reprogramming coordinates developmental and stress-responsive networks and integrates multiple signaling pathways—including photoperiod, temperature, gibberellin, and aging pathways—that converge on floral integrator genes to initiate inflorescence formation.

Several essential genes that govern dormancy signaling pathways have been found, including *DORMANCY-ASSOCIATED MADS-BOX (DAM)* genes and flowering-related genes such as *FLOWERING LOCUS T* (*FT*) and *SUPPRESSOR OF OVEREXPRESSION OF CONSTANS 1* (*SOC1*) (Böhlenius et al., 2006; Rinne et al., 2011; Vergara et al., 2016; Wang et al., 2020). Flowering is triggered by variations in the levels of protein-coding transcripts that either facilitate or suppress flowering (Achard et al., 2006). *FLOWERING LOCUS C* (*FLC*) transcripts impede

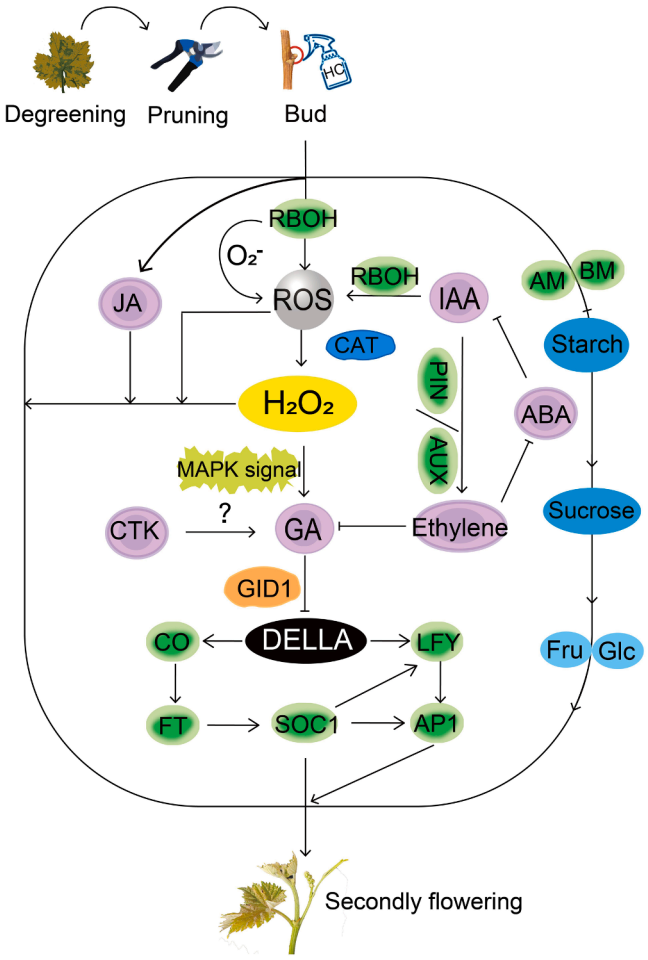


Fig. 1. Schematic representation of HC-mediated regulation of secondary flowering in grapevine.

flowering, while *FLOWERING LOCUS T* (*FT*), and *CONSTANS* (*CO*) promote it (Kramer and Hall, 2005). HC treatment promotes the release of grapevine buds from endodormancy by upregulating the expression of *VvFT, VvCO* and phytochrome genes (*VvPHYA/B*), while coordinately modulating the expression of repressor genes (*VvFLC, VvEBS*), which may indirectly influence flowering (Pérez et al., 2011). HC transiently activates hypoxia-responsive genes (*HRGs*) and *VvFT*, influencing bud metabolism and developmental processes, ultimately promoting bud break. These genes exhibit a rapid upregulation within 24 h post-treatment, followed by a decline. Prior to bud break, they are upregulated again, this time as part of developmental regulation rather than stress response, ensuring bud emergence at an optimal timing. This study elucidates the molecular mechanism by which stress (HC) and developmental signals (*VvFT*) collaboratively regulate dormancy release (Vergara et al., 2012).

VvFT and *VvMADS8* exhibit strong homology to *FT* and *SOC1* in *Arabidopsis*, respectively, and overexpression of either gene significantly accelerates flowering in *Arabidopsis* (Sreekantan and Thomas, 2006). In *Arabidopsis, CO* activates *SOC1* through *FT*, and *SOC1* directly regulates *LFY* (Lee and Lee, 2010). A comparable regulatory system may be present in grapevine, wherein the *SOC1* homolog *VvMADS8* presumably functions downstream of the *FT* homolog *VvFT* and directly modulates *LFY* (Sreekantan and Thomas, 2006).In *Arabidopsis, AP1* is a direct target of *LFY* and operates concurrently with it (Lee and Lee, 2010). Research indicates that *LFY* and *AP1* collaboratively modify transcription to govern the floral transition and first stages of flower development (Winter et al., 2015). Zhang et al. sequenced important grapevine floral meristem genes (*AP1, FT*, and *LFY* homologs). All species exhibited the

AP1 homolog only in tendrils and inflorescences, but the *LFY* homolog was only found in inflorescences. This shows that the *LFY* homolog may play a vital role in determining whether an uncommitted primordium develops into an inflorescence or a tendril; without it, tendrils are unable to produce floral organs (Zhang et al., 2015).

During the floral bud differentiation process following HC-induced bud break in winter buds of 'Shine Muscat' grapevine, *VvSOC1* and *VvFT* were markedly increased during the initial phase of bud differentiation, suggesting their role in the commencement of floral bud differentiation. *VvLFY*, and *VvAP1* exhibited elevated expression levels from the primordium formation stage to the inflorescence meristem differentiation stage, indicating their involvement in facilitating the growth of inflorescence meristems (Zhu, 2020).

3. Regulation of grapevine flowering for sustainable double-cropping cultivation

Comparatively, ethephon-induced leaf degreening accelerates leaf senescence and nutrient backflow but must be carefully managed to avoid excessive defoliation and reduced photosynthetic capacity. Pruning provides mechanical stimulation that enhances reactive oxygen species (ROS) signaling and promotes floral differentiation, though improper timing or intensity may increase susceptibility to pathogen invasion. HC application remains the most effective approach for synchronized budbreak and flowering; however, its narrow effective temperature window, potential phytotoxicity, and environmental risks associated with cyanamide residues limit its broader use (Ionescu et al., 2017; Liang et al., 2019; Zhang et al., 2023). In subtropical forcing-cultivation systems, where precise control of developmental time is needed to accomplish two grape harvest seasons per year without overlapping cultivation, such stress-induced regulation of budbreak and flowering is essential.

A survey of current literature and production practices shows that similar strategies have been applied across a range of perennial fruit crops, including blueberry, tea oil *camellia*, sweet cherry, and grapevine, with reported outcomes such as enhanced floral bud break, advanced flowering, and improved ripening. These practices range from experimental trials to commercial applications, highlighting the broader relevance and potential of cultivation-based interventions in crop phenology regulation (Table 1).

3.1. Leaf degreening

Grapevines are deciduous fruit trees, but in southern regions, due to climatic limitations and insufficient chill hours, they struggle to naturally enter dormancy (Anzanello et al., 2018). This substantially influences the process of floral development for the second cropping cycle. To initiate dormancy in grapevines following the summer harvest for the subsequent crop, the use of suitable degreening chemicals is essential. The use of degreening chemicals like ethephon following the summer fruit harvest facilitates leaf degreening, nutrient backflow, and dormancy simulation, all of which are critical for enhancing budburst and secondary flower bud differentiation (Qiu et al., 2019). Ethephon treatment in *Vitis vinifera* 'Riesling' induced premature leaf yellowing and defoliation, accompanied by elevated expression of *WRKY53, senescence-associated gene 13* (*SAG13*), and *glutamate dehydrogenase* (*GDH*), indicating an ethylene-induced activation of leaf senescence (Hendgen et al., 2021).

The application of leaf degreening treatment accelerates grapevine budburst and blooming by 3–5 days (Yang et al., 2021). This effect is attributed to the rapid degradation of chloroplast pigments (chlorophyll a/b, carotenoids), which inhibits photosynthesis and initiates leaf senescence through increased malondialdehyde (MDA) levels and depletion of starch and sugars. Additionally, this process promotes the remobilization of nutrients from ageing leaves to branches, ensuring preservation within a three-week period (Yang et al., 2021). About seven days after vine recovery (approximately 15 days post-harvest), 40 % ethephon and 80 % sulfur were sequentially applied twice at 15-day intervals, followed by irrigation withholding until five days before summer pruning, which significantly enhanced shoot maturation and nutrient backflow in 'Summer Black' grapes and induced leaf yellowing and abscission within approximately 25 days (Qiu et al., 2019).

3.2. Pruning the bearing base shoot

Pruning significantly influences the growth structure, yield, fruit quality, and the balance between vegetative and reproductive growth of grapevines (Bertamini and Faralli, 2023; Fazeli Rostampour, 2022). Pruning serves as an abiotic stressor in grapes, influencing flowering by either inducing or accelerating it or conversely inhibiting or delaying it. Transcriptome analysis revealed that pruning triggers ROS accumulation and activates wound-response and ethylene signaling, which are amplified via the MAPK signaling pathway to regulate downstream genes involved in floral bud differentiation and growth (Fan et al., 2023). This effect is contingent upon the timing, local climatic conditions, and cultivation practices employed (Petrie et al., 2017; Trought et al., 2011; Zhang et al., 2023). The placement and diameter of the cut branches are vital for the growth of secondary flowers of winter buds, and various types choose them differently (Shanmugasundram et al., 2015). Vegetative and reproductive growth are regulated by modifying the length of bearing mother shoots, where high-yielding varieties are pruned to shorter shoots, while low-yielding varieties are pruned to medium-long or long shoots (El-kereamy and Kurtural, 2022).

Green shoot pruning and lateral shoot trimming were conducted at node six beginning in mid-May for the second crop production. This method facilitated bud break at the specified nodes and the subsequent development of blooming shoots. Each fruiting shoot bore a single cluster comprising up to 60 berries (Fan et al., 2023). Pruning the main shoots by retaining the basal six nodes and removing the apical meristem stimulates shoot regeneration and enables the production of a delayed second-season crop. This pruning technique also promotes the formation of inflorescences bearing a greater number of flowers (Poni et al., 2020).

Pruning at appropriate nodes significantly influences budburst, floral differentiation, and fruit set, which are critical for achieving successful two-cropping. The optimal number and position of retained nodes are

Table 1
Stress-induced flowering applications in different crops.

Crop Variety	Method	Application Rate	Observed Effect	Application Context	References
Blueberry	HC	0.67 % spray	Promoted floral bud break (76 %)	Experimental field trials	(Wang et al., 2021)
Tea oil *Camellia*	HC	0.5 % spray	Flowering advanced 12 d	Experimental field trials	(Lin et al., 2022)
Sweet Cherry	HC	2 % spray	Promoted floral bud break (97 %)	Experimental field trials	(Ionescu et al., 2017)
Grape	HC	2.5 % spray	Promoted floral bud break (97 %)	Experimental field trials	(Liang et al., 2019)
Grape	Pruning and HC	Pruning at 8–10 nodes + 2.5 % HC spray	Promoted floral bud break (72 %) and uniform budbreak	Commercialized protected cultivation	(Qiu et al., 2019)

variety-specific. Intermediate base vine retained6–10 nodes enhances budburst rates (up to 80 %), facilitates floral differentiation, and secures successful fruit set for the second crop, thus fulfilling the objective of two-cropping (Huang and Li, 2017). In the imported grape varieties 'Jingyou', '8612′, and 'Wink' from Guangxi, China, pruning with a diameter exceeding 0.8 cm resulted in a flowering rate of over 80 % (Zhao et al., 2013). In double-cropping of 'Summer Black' grapes, retaining 4–8 winter buds and 7–10 summer buds markedly increased germination and flowering rates compared with lower bud numbers (Qiu et al., 2019). Apical pruning at nodes 9–13 in 'Kyoho' grapevines efficiently regulates vegetative vigor and significantly enhances budbreak and floral meristem differentiation efficiency (Li et al., 2013).

3.3. Dormancy-breaking with hydrogen cyanamide

In regions characterized by mild winters, the synergistic application of dormancy-breaking agents, including hydrogen cyanamide (HC), nitrogen-based compounds (e.g., calcium cyanamide), mineral oil, garlic extract, and hydrogen peroxide (H_2O_2), in conjunction with low-chill cultivars, significantly enhances the induction of flowering and fruit set (Leonel et al., 2015; Melke, 2015). Among these agents, hydrogen cyanamide (HC) has become the preferred compound for dormancy release in grape production. This preference arises from its ability to closely mimic the physiological signaling processes of natural dormancy release, producing consistent and reliable results across cultivars and environments. Moreover, HC offers practical advantages, including cost-effectiveness, ease of application, and a well-established record of efficacy in commercial viticulture. Consequently, HC is extensively applied to satisfy chilling requirements, promote uniform budbreak, and enhance floral bud differentiation, thereby facilitating both germination and secondary flowering (Chen et al., 2016; Jamshidian et al., 2024; Vergara et al., 2012). Hydrogen cyanamide (HC) promotes budbreak and flowering in grapevines, partly by enhancing reactive oxygen species (ROS) signaling and modulating carbohydrate and hormonal balance, thereby releasing dormancy and activating flowering genes (Fan et al., 2023; Li et al., 2024; Zhu, 2020). The degradation of HC in treated flower buds produces nitroxyl (HNO) and cyanide (CN^-), which markedly accelerates flowering time in *Vitis vinifera* (grape), *Prunus persica* (peach), *Diospyros kaki* (persimmon), *Prunus armeniaca* (apricot) *Malus domestica* (apple), and *Prunus avium* (sweet cherry) relative to untreated controls (Faust et al., 1997; Qiu et al., 2019; Sánchez-Pérez et al., 2014).

Under pruning conditions, the influence of HC on stimulating budbreak and floral bud differentiation is significantly amplified (Sudawan et al., 2016). In 5-year-old blueberry plants, HC treatment at various concentrations advanced the budbreak phase by 17–19 days, the flowering phase by 2–3 days, and the fruit ripening phase by 6–7 days (Wang et al., 2021). The summer grape crop exhibited longer intervals from HC application to budburst and full flowering, as well as from full flowering to fruit veraison, but a shorter veraison-to-ripening duration compared to the winter grape crop. These differences are primarily attributed to lower temperatures during flower development and fruit enlargement, and higher temperatures during ripening (Qiu et al., 2019). Twenty days post-treatment, grape buds treated with HC exhibited enlarged shoot apices, increased cells, well-developed inflorescence primordia, and intact floral meristems, in contrast to the water-treated controls (Tan et al., 2021).

3.4. The yield of the summer crop affects flower sprouting of the winter crop

In double-cropping systems, yield regulation represents a critical trade-off between productivity and resource allocation. For instance, in 'BRS Isis' seedless grape, summer yield reached 49 t ha^{-1}, whereas load adjustment in the off-season reduced yield to 24 t ha^{-1}, achieving a more balanced production (Ahmed et al., 2019). In a double-cropping system, summer grape yield is significantly negatively correlated with winter flowering rate; to ensure sustainability, the winter-to-summer harvest ratio should be maintained between 2:5 and 3:5 (Qiu et al., 2019). Moderate crop load regulation (e.g., bunch thinning), although reducing total yield, helps to alleviate source–sink competition and improve fruit quality traits, including increased anthocyanin concentration and soluble solids content (Avizcuri-Inac et al., 2013; Gil et al., 2013). Therefore, appropriate load management not only contributes to maintaining annual yield balance but also optimizes fruit composition, achieving an optimal trade-off between yield and quality under double-cropping viticulture systems.

4. Strategies for selecting grapevine varieties for double cropping

4.1. Double cropping models: emphasis on the non-overlapping system

This planting model is primarily implemented through two distinct approaches (Fig. 2). The first of model is the "two generations under one roof", where the first and second harvests overlap, resulting in the simultaneous presence of fruits and flowers on the same plant or bearing branch. This model presents significant challenges, as balancing water and nutrient allocation between developing fruits and flowers is difficult. Additionally, the simultaneous management of pests and diseases affecting both flowers and fruits complicates cultivation, often leading to compromised berry quality (Martínez-Moreno et al., 2019; Mitra et al., 2018).

The second model, referred to as the "two generations different" pattern, entails the harvesting of grapes throughout two distinct, non-concurrent seasons—two blooms (primary and secondary), specifically in summer and winter.

This approach alleviates many of the limitations of the overlapping model by enabling more balanced allocation of water, nutrients, and assimilates across distinct growth cycles. Physiologically, the non-overlapping schedule allows vines to recover vegetative vigor and

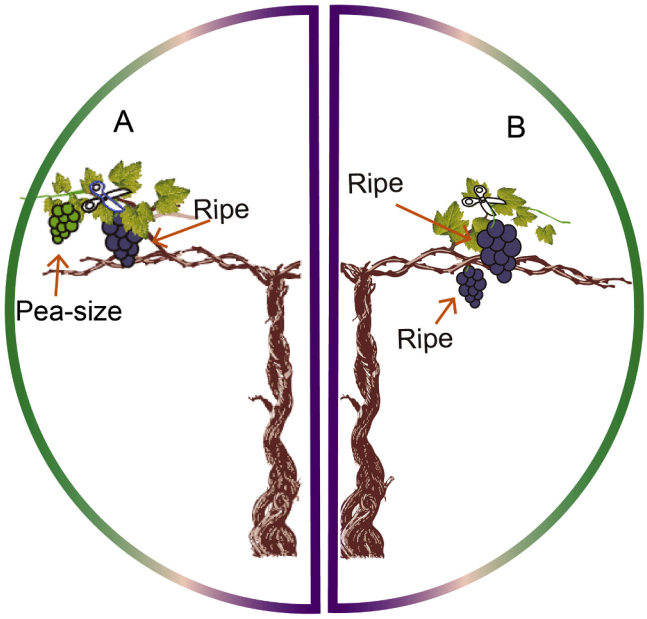


Fig. 2. Two main cultivation modes of grapevine double cropping.
(A) The " two generations under one roof" model: this model features pea-sized young fruits and ripe fruits coexisting on the same plant or branch, with staggered maturity periods. (B) The "two different generations" model (summer and winter): the dotted lines in the illustration represent the harvesting process of grapes across two distinct and non-overlapping seasons. Grapes are harvested in two distinct seasons: summer (April to September) and winter (October to March), with no overlap between the two periods.

replenish carbohydrate reserves after the summer harvest, reducing inter-seasonal competition for photosynthates and hormonal signals. This restoration phase enhances bud differentiation and flowering uniformity in the subsequent cycle (Bai et al., 2008; Qiu et al., 2019). Following the summer grape harvest, a period of over a month is dedicated to the implementation of rational nutrient adjustment, with the objective of restoring vine vigour. Subsequently, chemicals are employed to induce leaf degreening, followed by pruning and targeted dormancy-breaking chemicals on vines that are dormant or semi-dormant, respectively. Trench fertilisation follows leaf degreening to guarantee timely water and nutrient availability. To prevent illnesses and pests, gardens are thoroughly sanitised and disinfected following trimming. When inflorescences emerge, tendrils are removed and clusters are thinned. These technical approaches encourage bud sprouting and secondary blooming in the same season, resulting in a new growth cycle and year-round grape production (Fig. 3).

4.2. Varietal selection

Currently, there are around 6000 grapevine cultivars available for protected growing globally (Díaz-Fernández et al., 2022). The successful implementation of this cultivation model is closely associated with the adaptability of grape cultivars, particularly their dormancy behaviour, temperature sensitivity, flowering plasticity, stress responsiveness, and maturation timing. For instance, cultivars with lower chilling requirements and strong regenerative capacity are better adapted to the "two generations different" model, as they can resume growth and reinitiate flowering after a short recovery period. In contrast, high-chill varieties are more constrained by overlapping schedules due to delayed bud break and incomplete dormancy release (Anzanello et al., 2018). Therefore, aligning varietal selection with the physiological demands of each cropping system is essential for maintaining yield stability and fruit quality across seasons.

Abiotic stress has a significant impact on grapevine blooming, with various cultivars showing distinct patterns in both stress responses and flower bud differentiation and fruiting behaviour (Chen et al., 2023). Under adverse environmental conditions such as drought or temperature extremes, grapevine cultivars exhibit distinct flowering plasticity driven by genotype–environment–hormone interactions (Dou et al., 2024). Hydrogen cyanamide (HC) modulates this process by coordinating GA, ABA, CTK, ethylene, and JA signaling—downregulating ABA biosynthesis (*VvNCED, VvXERICO*) while enhancing GA- and ethylene-related gene expression (*VvGA3ox, VvACS, VvACO*)—to promote dormancy release and bud activation (Liang et al., 2019; Lin and Agehara, 2021; Li et al., 2024). Genotypic variation in hormonal responsiveness determines the timing and stability of flowering, providing a physiological basis for cultivar adaptability under double-cropping systems. The major consideration while employing the secondary fruiting of grapes should be their maturation duration. The varieties in Table 2 have been recognized as especially appropriate for the implementation of non-overlapping two-crop systems. Early-maturing cultivars, including 'Summer Black', 'Kyoho', 'Pinot Noir', 'Muscat Hamburg', 'Victoria', 'Hu Tai 8′, 'Zui Jin Xiang', and 'Golden Finger', exhibit notable efficacy under abiotic stressors, such as hydrogen cyanamide spraying and pruning (Bertamini and Faralli, 2023; Chen et al., 2021; Cheng et al., 2019; Deng et al., 2020; Li et al., 2022; Lu et al., 2021; Poni et al., 2020; Yao et al., 2021; Zha et al., 2017). Early flowering, great quality, early ripening, high humidity resistance, and low chilling requirements characterize these types (Guo et al., 2016). Conversely, late-maturing varieties such as 'Shine Muscat' and 'Syrah' demonstrate greater suitability for postponed flowering and harvesting (Pedro et al., 2017; Wang et al., 2022).

Integrating practices such as pruning, the application of HC, and the careful selection of varieties with staggered harvest times enables vineyards to achieve year-round production of table grapes. The growth and developmental physiology of grape varieties also contribute significantly. Fig. 4 depicts the seasonal production trends of principal grape types. Early-maturing cultivars, such as 'Summer Black', yield fruit from April to July and again from November to January, so facilitating dual harvests in summer and winter. Mid- and late-maturing cultivars, such as 'Kyoho' and 'Shine Muscat,' are systematically arranged to address output deficits in February, March, August, and September. Specifically, 'Summer Black' enhances the early summer market, whilst 'Shine Muscat' produces high-value fruit in June, July, and December. Conversely, 'Kyoho' is primarily gathered in the winter and early spring months, specifically from November to March, so providing a consistent supply (Bai et al., 2016; Chen, 2023; Pan et al., 2022; Qiu et al., 2019; Wang et al., 2024). This strategic integration guarantees a consistent supply of fresh grapes to satisfy market demand, while upholding a stable and sustainable production system throughout the year.

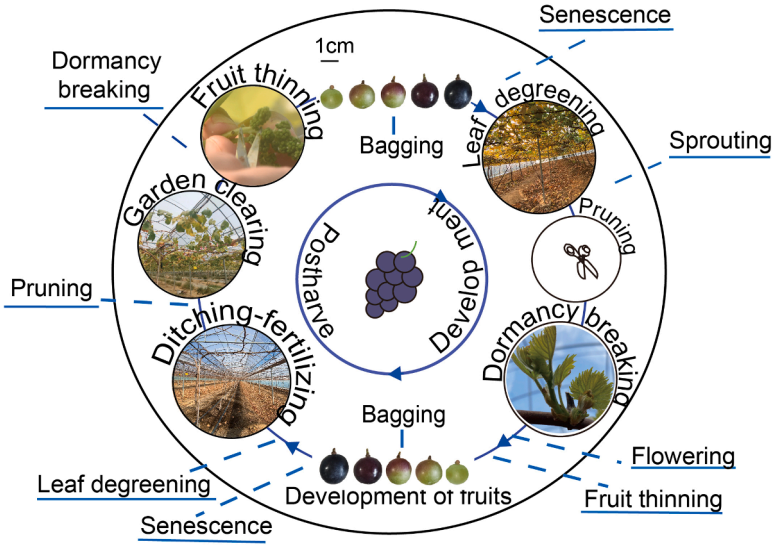


Fig. 3. Grapevine double cropping without overlapping.
Note: The time from full flowering to fruit ripening in the summer crop is longer than that in the winter crop.

Table 2
Grape varieties suitable for two-cropping cultivation.

Variety	Characteristic	Harvest (Summer)	Harvest (Winter)	Climate zone	References
Summer Black	Early-maturing	Apr-May*, Jun-Jul	Oct-Nov, Dec-Jan	Subtropical	(Chen et al., 2021; Li et al., 2023; Qiu et al., 2019)
Shine Muscat	Late-maturing	Jun-Jul*, Aug-Sept	Dec-Jan, Feb-Mar	Subtropical	(Pan et al., 2022)
Kyoho	Early-maturing	Jun-Jul	Dec-Jan	Tropical	(Cheng et al., 2019)
ZuiJinXiang	Early-maturing	Jul-Aug	Dec-Jan	Subtropical	(Zha et al., 2017)
Gold Finger	Early-maturing	Jul-Aug	Dec-Jan	Subtropical	(Deng et al., 2020)
Hu Tai 8	Early-maturing	Jul-Aug	Dec-Jan	Subtropical	(Li et al., 2022)
Pinot Noir	Early-maturing	Sept-Oct	Feb-Mar	Temperate	(Bertamini and Faralli, 2023)
Muscat Hamburg	Early-maturing	Jul-Aug	Dec-Jan	Tropical	(Lu et al., 2021)
Victoria	Early-maturing	Jul-Aug	Dec-Jan	Tropical	(Lu et al., 2021)
Syrah	Late-maturing	Jul-Aug	Jan-Feb	Subtropical	(Pedro et al., 2017)

Note: Through the promotion of early-maturing cultivation. 'Summer Black' grapes are picked in April and May, and 'Shine Muscat' grapes are harvested in June and July.

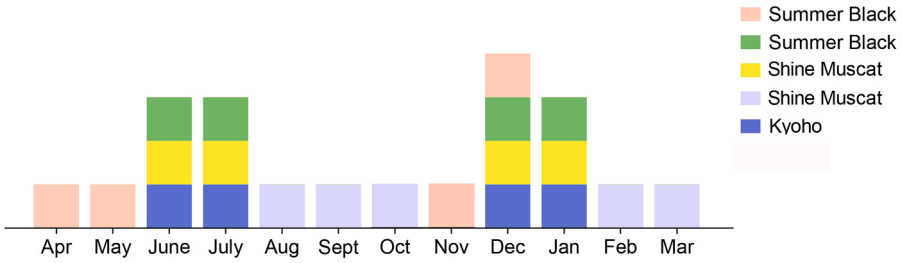


Fig. 4. Grape year-round supply in Grapevine yard, China.
Note: The same color reflects the first and second grape harvesting periods.

5. Challenges and prospects

5.1. Challenge

Leaf degreening, pruning, and hydrogen cyanamide (HC) are widely used to regulate flowering in grapevine by modulating sugar, hormone, and carbohydrate pathways. Leaf degreening promotes floral induction through the accumulation of metabolites such as abscisic acid and cytokinin, while pruning redistributes carbohydrates and nutrients, reallocating resources to developing buds and activating wound- and reactive oxygen species (ROS)-related signaling. HC treatment further triggers broad-scale regulation of genes related to redox, hormone, and carbohydrate pathways, including the activation of biosynthetic enzymes for gibberellins and cytokinins, while coordinating their expression during dormancy release and flowering through epigenetic mechanisms (Chen et al., 2025; De Rosa et al., 2023; Han et al., 2024; Li et al., 2024).However, how these molecular pathways integrate stress responses with developmental regulation across consecutive cropping cycles, and how field practices can be optimized based on these molecular insights, constitute major research gaps. Addressing these challenges is essential for improving the efficiency and sustainability of double-cropping systems in grapevine.

The practical challenges of hydrogen cyanamide (HC) application are primarily related to dosage and environmental conditions. HC's effectiveness in breaking dormancy is concentration-dependent: low doses are ineffective, while high concentrations can cause reactive oxygen species (ROS) accumulation and bud damage. For instance, concentrations below 1.5 % show limited efficacy, while those above 2.5 % result in significant damage and reduced flowering (Arora et al., 2011). Seasonal variability, particularly temperature fluctuations, further affects HC efficacy. In cooler climates, such as parts of New Zealand, low winter temperatures reduce HC absorption, while warmer temperatures in late spring can cause premature bud break if applied too early, leading to suboptimal flowering and fruit set (Schep et al., 2009). In addition to these agronomic challenges, improper handling and excessive exposure to hydrogen cyanamide (HC) may pose health risks to humans. Inhalation or dermal contact with HC can induce symptoms such as headaches, and in severe cases, may result in respiratory distress and neurological symptoms due to oxidative stress (Devi, 2021).

5.2. Prospect

Regulating flowering is essential for grape production. The conventional single-harvesting model has struggled to satisfy contemporary market demands, which require a breakthrough in innovative cultivation strategies. Currently, leaf degreening and pruning, along with HC application, effectively disrupt the dormant period, promote orderly bud germination, and induce secondary flowering, thereby facilitating a continuous production cycle. Future advancements in flowering induction strategies will be essential for enhancing viticulture's sustainability and economic viability. Precision viticulture tools, such as remote sensing, multispectral imaging, and machine-learning-based canopy or phenology modeling, can be employed to monitor bud status, dormancy progression, and environmental micro-conditions in real time (Canavera et al., 2023; Mucalo et al., 2024). These technologies will allow site-specific management of pruning intensity, HC dosage, and application timing, thereby minimizing phytotoxicity risks and optimizing resource use.

At the same time, the development of alternative dormancy-breaking agents is becoming increasingly important. Environmentally benign compounds—such as hydrogen peroxide–based formulations, thiourea derivatives, or naturally derived signaling molecules (e.g., nitric oxide donors, amino acid chelators)—offer promising avenues to replace or complement HC in sustainable production systems (Mujahid et al., 2020; Venter et al., 2024). Combining these with antioxidant modulators or biostimulants may improve bud vitality while reducing environmental and health concerns. In parallel, breeding and molecular design of low-chill or temperature-adaptive cultivars will be a long-term solution to reduce dependency on chemical treatments. Overall, these approaches will pave the way toward a resilient and eco-efficient grape production system characterized by consistent flowering, stable yields, and superior fruit quality across diverse climatic regions.

Actionable research priorities include: (i) integrating multi-omics data to elucidate the molecular networks linking pruning, leaf

degreening, and dormancy release; (ii) developing predictive models for precision regulation of bud break under variable climates; and (iii) screening and validating low-toxicity alternatives to HC for environmentally sustainable viticulture.

CRediT authorship contribution statement

Chengyue Li: Writing – original draft, Conceptualization. **Congqiao Wang:** Methodology, Data curation, Conceptualization. **Zhipeng Qiu:** Resources. **Youhuan Li:** Software. **Yue Zeng:** Data curation. **Siyu Yang:** Formal analysis. **Cuilan Ma:** Formal analysis. **Dongliang Qiu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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